

CLINICAL PRESENTATION	DIAGNOSIS/DISEASE	PAGE
Episodic vertigo, tinnitus, sensorineural hearing loss	Ménière disease (↑ endolymph in inner ear)	548
Ptosis, miosis, anhidrosis	Horner syndrome (sympathetic chain lesion)	555
Conjugate horizontal gaze palsy, horizontal diplopia	Internuclear ophthalmoplegia (MLF lesion; unilateral or bilateral)	558
Acute onset, reversible “waxing and waning” level of consciousness (acute onset), ↓ attention span, ↓ arousal	Delirium (usually 2° to other cause)	575
Polyuria, renal tubular acidosis type II, growth retardation, electrolyte imbalances, hypophosphatemic rickets	Fanconi syndrome (generalized reabsorption defect of proximal convoluted tubule)	604, 611
Periorbital and/or peripheral edema, proteinuria (> 3.5 g/day; frothy urine), hypoalbuminemia, hypercholesterolemia	Nephrotic syndrome (podocyte damage)	613
Hereditary nephritis, sensorineural hearing loss, retinopathy, anterior lenticonus	Alport syndrome (type IV collagen mutation)	615
Wilms tumor, macroglossia, organomegaly, hemihyperplasia, omphalocele	Beckwith-Wiedemann syndrome (WT2 mutation)	624
Streak ovaries, congenital heart disease, horseshoe kidney, cystic hygroma, short stature, webbed neck, lymphedema	Turner syndrome (45,XO)	655
Ovarian fibroma, ascites, pleural effusion	Meigs syndrome	665
Red, itchy, swollen rash of nipple/areola	Paget disease of the breast (sign of underlying neoplasm)	668
Fibrous plaques in tunica albuginea of penis with abnormal curvature	Peyronie disease (associated with repeated minor trauma during intercourse)	669
Hypoxemia, polycythemia, hypercapnia	Chronic bronchitis (hypertrophy and hyperplasia of mucus-secreting glands)	692
Pink complexion, dyspnea, hyperventilation	Emphysema (centriacinar [tobacco smoking] or panacinar [α_1 -antitrypsin deficiency])	692
Bilateral hilar adenopathy, uveitis, arthropathy, skin changes	Sarcoidosis (immune-mediated noncaseating granulomas)	695

► CLASSIC LABS/FINDINGS

LAB/DIAGNOSTIC FINDING	DIAGNOSIS/DISEASE	PAGE
Colonies of <i>Pseudomonas</i> in lungs, ↑ Cl^- on sweat test, ↑ immunoreactive trypsinogen	Cystic fibrosis (autosomal recessive mutation in <i>CFTR</i> gene → fat-soluble vitamin deficiency and mucous plugs)	58
↓ AFP on second trimester screening	Down syndrome, Edwards syndrome	61
↑ β -hCG, ↓ PAPP-A on first trimester screening	Down syndrome	61
Homocysteine, methylmalonic acid, folate	Vitamin B ₁₂ deficiency	67
Anti-histone antibodies	Drug-induced lupus	113
↓ T cells, ↓ PTH, ↓ Ca^{2+} , absent thymic shadow on CXR	Thymic aplasia (22q11 microdeletion: DiGeorge syndrome, velocardiofacial syndrome)	114
Recurrent infections, eczema, thrombocytopenia, ↑ IgE and IgA, ↓/normal IgG and IgM	Wiskott-Aldrich syndrome (WAS gene mutation)	115

LAB/DIAGNOSTIC FINDING	DIAGNOSIS/DISEASE	PAGE
Giant granules in phagocytes, pancytopenia (immunodeficiency)	Chédiak-Higashi disease (<i>LYST</i> gene mutation: congenital failure of phagolysosome formation)	115
Optochin sensitivity	Sensitive: <i>S pneumoniae</i> ; resistant: viridans streptococci (<i>S mutans</i> , <i>S sanguis</i> , <i>S mitis</i>)	132
Novobiocin response	Sensitive: <i>S epidermidis</i> ; resistant: <i>S saprophyticus</i>	132
Bacitracin response	Sensitive: <i>S pyogenes</i> (group A); resistant: <i>S agalactiae</i> (group B)	132
Branching gram ⊕ rods with “sulfur granules”	<i>Actinomyces israelii</i>	137
Hilar lymphadenopathy, peripheral granulomatous lesion in middle or lower lung lobes (can calcify)	Ghon complex (1° TB: <i>Mycobacterium bacilli</i>)	138
“Thumb sign” on lateral neck x-ray	Epiglottitis (<i>Haemophilus influenzae</i>)	140
Bacteria-covered vaginal epithelial cells, ⊕ whiff test	“Clue cells” (<i>Gardnerella vaginalis</i>)	147
Ring-enhancing brain lesion on CT/MRI in AIDS	<i>Toxoplasma gondii</i> (multiple), CNS lymphoma (may be solitary)	153, 173
Dilated cardiomyopathy with apical atrophy, megacolon, megaesophagus	Chagas disease (<i>Trypanosoma cruzi</i>)	155
Atypical lymphocytes, heterophile antibodies	Infectious mononucleosis (EBV infection)	162
Narrowing of upper trachea and subglottis (Steeple sign) on x-ray	Croup (parainfluenza virus)	167
Cytoplasmic inclusions in Purkinje cells of cerebellum and in hippocampal neurons	Negri bodies of rabies	169
Concentrically laminated calcified spherules (psammoma bodies)	Papillary thyroid carcinoma, somatostatinoma, meningioma, mesothelioma, ovarian serous papillary cystadenocarcinoma, prolactinoma	207
Boot-shaped heart on x-ray	Tetralogy of Fallot (due to RVH)	302
Rib notching (inferior surface, on x-ray)	Coarctation of the aorta	304
Delta wave and shortened PR interval on ECG; may lead to supraventricular tachycardia	Wolff-Parkinson-White syndrome (bundle of Kent bypasses AV node)	311
Electrical alternans (alternating amplitude of R waves on ECG)	Cardiac tamponade	317
Granuloma with giant cells after pharyngeal infection	Aschoff bodies (rheumatic fever)	319
↑ level of fibrin degradation products (D-dimer)	DVT, DIC	321, 433
Empty-appearing nuclei with central clearing in thyroid cells	“Orphan Annie” eyes (papillary thyroid carcinoma)	347
“Brown” tumor of bone	Hyperparathyroidism or osteitis fibrosa cystica (deposited hemosiderin from hemorrhages causes brown color)	349, 469
Hypertension, hypokalemia, metabolic alkalosis, ↑ aldosterone, ↓ renin	1° hyperaldosteronism (eg, Conn syndrome)	354
Mucin-filled cell with peripheral nucleus	“Signet ring” cells (diffuse gastric carcinoma)	386
Anti-transglutaminase/anti-deamidated gliadin/anti-endomysial antibodies	Celiac disease (diarrhea, weight loss)	388
Narrowing of bowel lumen on small bowel follow-through	“String sign” (Crohn disease)	389

LAB/DIAGNOSTIC FINDING	DIAGNOSIS/DISEASE	PAGE
“Lead pipe” appearance of colon on imaging	Ulcerative colitis (loss of haustra)	389
Thousands of polyps on colonoscopy after puberty	Familial adenomatous polyposis (autosomal dominant APC gene mutation)	394
“Apple core” lesion on barium enema x-ray	Colorectal cancer (usually left-sided)	395
“Nutmeg” (mottled) appearance of liver	Chronic passive congestion of liver due to right heart failure, Budd-Chiari syndrome	397
Eosinophilic cytoplasmic inclusion of damaged keratin within hepatocyte	Mallory body (alcoholic hepatitis)	398
Fatty infiltration of hepatocytes → cellular “ballooning”, eventual necrosis	Fatty liver disease (associated with alcohol use or metabolic syndrome)	398
Anti-smooth muscle antibodies (ASMAs), anti-liver/kidney microsomal-1 (anti-LKM1) antibodies	Autoimmune hepatitis	399
Antimitochondrial antibodies (AMAs)	1° biliary cholangitis (female, cholestasis, portal hypertension)	402
↓ serum ceruloplasmin	Wilson disease	402
Migratory thrombophlebitis (leading to migrating DVTs and vasculitis)	Trousseau syndrome (pancreatic adenocarcinoma)	405
Basophilic stippling of RBCs (basophilic ribosomal precipitates)	Sideroblastic anemia, thalassemias	421
Basophilic nuclear remnants in RBCs	Howell-Jolly bodies (due to splenectomy or nonfunctional spleen)	421
Hypochromic, microcytic anemia	Iron deficiency anemia, lead poisoning, thalassemia (fetal hemoglobin sometimes present), sideroblastic anemia	424, 425
“Hair on end” (“crew cut”) appearance on skull x-ray	β-thalassemia, sickle cell anemia (marrow expansion)	425, 428
Hypersegmented neutrophils	Megaloblastic anemia (vitamin B ₁₂ deficiency: neurologic symptoms; folate deficiency: no neurologic symptoms)	426
Anti-GpIIb/IIIa antibodies	Immune thrombocytopenia	432
Giant B cells featuring bilobed nuclei with prominent inclusions (“owl’s eye”)	Reed-Sternberg cells (Hodgkin lymphoma)	434
Sheets of lymphocytes with interspersed “tingible body” macrophages (“starry sky” histology)	Burkitt lymphoma [t(8:14) c-myc activation, associated with EBV; “starry sky” made up of malignant cells]	435
Lytic (“punched-out”) bone lesions on x-ray	Multiple myeloma	436
Monoclonal spike on serum protein electrophoresis	Multiple myeloma (usually IgG or IgA); Waldenström macroglobulinemia (IgM); monoclonal gammopathy of undetermined significance	436
Stacks of RBCs	Rouleaux formation (↑ ESR, multiple myeloma)	436
Myeloperoxidase ⊕ cytoplasmic inclusions in myeloblasts, ↑↑↑ circulating myeloblasts	Auer rods (APL)	437
WBCs that look “smudged”	CLL	437
“Tennis racket”-shaped cytoplasmic organelles (on EM) in Langerhans cells	Birbeck granules (Langerhans cell histiocytosis)	439

LAB/DIAGNOSTIC FINDING	DIAGNOSIS/DISEASE	PAGE
“Soap bubble” in femur or tibia on x-ray	Giant cell tumor of bone (generally benign)	470
Elevated periosteum (creating a “Codman triangle”) or sunburst pattern on x-ray	Osteosarcoma (osteogenic sarcoma)	471
“Onion skin” periosteal reaction	Ewing sarcoma (malignant small blue cell tumor)	471
IgM antibodies that target IgG Fc region, anti-cyclic citrullinated peptide antibodies	Rheumatoid arthritis (systemic inflammation, joint pannus, boutonniere and swan neck deformities)	472
Needle-shaped, $\ominus$ birefringent crystals	Gout (monosodium urate crystals)	473
$\uparrow$ uric acid levels	Gout, Lesch-Nyhan syndrome, tumor lysis syndrome, loop and thiazide diuretics	473
Rhomboid, $\oplus$ birefringent crystals	Pseudogout (calcium pyrophosphate dihydrate crystals)	473
“Bamboo spine” (vertebral fusion) on x-ray	Ankylosing spondylitis (chronic inflammatory arthritis, associated with HLA-B27)	475
Antinuclear, anti-Smith, anti-dsDNA, antiphospholipid antibodies	SLE (type III, and to a lesser degree, type II hypersensitivity)	476
Antineutrophil cytoplasmic antibodies (ANCA)	Microscopic polyangiitis, eosinophilic granulomatosis with polyangiitis, and primary sclerosing cholangitis (MPO-ANCA/p-ANCA); granulomatosis with polyangiitis (PR3-ANCA/c-ANCA)	402, 478
Anti-Scl-70 (anti-DNA topoisomerase-I) and anti-RNA polymerase III antibodies	Diffuse scleroderma	481
Anti-centromere antibodies	Limited scleroderma (CREST syndrome)	481
Anti-desmoglein 3 $\pm$ 1 (anti-desmosome) antibodies	Pemphigus vulgaris	489
Anti-hemidesmosome antibodies	Bullous pemphigoid	489
Keratin pearls on skin biopsy	Squamous cell carcinoma	494
$\uparrow$ AFP in maternal serum	Dating error, open neural tube defects	501
Bloody or yellow CSF on lumbar puncture	Xanthochromia (due to subarachnoid hemorrhage)	528
Loss of dopaminergic (pigmented) neurons in substantia nigra	Parkinson disease	534
Intracellular eosinophilic inclusion in neuron	Lewy body, composed of α -synuclein (Parkinson disease and Lewy body dementia)	534, 535
Extracellular amyloid deposition in gray matter of brain	Senile plaques (Alzheimer disease)	534
Hyperphosphorylated tau protein aggregates in neurons	Neurofibrillary tangles (Alzheimer disease) and Pick bodies (frontotemporal dementia)	534, 535
“Pseudopalisading” pleomorphic tumor cells on brain biopsy	Glioblastoma	540
Small blue cells surrounding central area of neuropil	Homer-Wright rosettes (neuroblastoma, medulloblastoma)	354, 542
RBC casts in urine	Glomerulonephritis, hypertensive emergency	612
WBC casts in urine	Acute pyelonephritis, transplant rejection, tubulointerstitial inflammation	612
Granular, “muddy-brown” casts in urine	Acute tubular necrosis (eg, ischemia or toxic injury)	612
Waxy casts with very low urine flow	End-stage renal disease/chronic kidney disease	612

LAB/DIAGNOSTIC FINDING	DIAGNOSIS/DISEASE	PAGE
“Lumpy-bumpy” appearance of glomeruli on immunofluorescence	Infection-related glomerulonephritis (due to IgG, IgM, and C3 deposition)	614
Anti-glomerular basement membrane antibodies, linear appearance of IgG deposition on glomerular and alveolar basement membranes	Goodpasture syndrome (hematuria, hemoptysis)	614
		614
Necrotizing vasculitis (lungs) and necrotizing glomerulonephritis	Granulomatosis with polyangiitis (PR3-ANCA/c-ANCA) and Goodpasture syndrome (anti-basement membrane antibodies)	614, 479
Cellular crescents in Bowman's space on light microscopy	Rapidly progressive (crescentic) glomerulonephritis	614
“Wire loop” glomerular capillary on light microscopy	Diffuse proliferative glomerulonephritis (usually seen with lupus)	614
“Tram-track” appearance of capillary loops of glomerular basement membranes on light microscopy	Membranoproliferative glomerulonephritis	615
Effacement of podocyte foot processes on EM	Minimal change disease (child with nephrotic syndrome), focal segmental glomerulosclerosis (nephrotic syndrome more common in Black individuals)	616
“Spike and dome” appearance of subepithelial deposits	Membranous nephropathy (nephrotic syndrome)	616
Eosinophilic nodular glomerulosclerosis	Kimmelstiel-Wilson lesions (diabetic glomerulonephropathy)	616
Thyroid-like appearance of kidney	Chronic pyelonephritis	619
↑ hCG	Multifetal gestation, hydatidiform mole, choriocarcinoma, dysgerminoma, Down syndrome	652
Dysplastic squamous cervical cells with “raisinoid” nuclei and perinuclear halo	Koilocytes (HPV infection; predisposes to cervical cancer)	663
Sheets of uniform “fried egg” cells, ↑ hCG, ↑ LDH	Dysgerminoma	664
Schiller-Duval bodies (resemble glomeruli), ↑ AFP	Yolk sac tumor	664
Disarrayed granulosa cells arranged around collections of eosinophilic fluid	Call-Exner bodies (granulosa cell tumor of the ovary)	665
“Chocolate cyst” in ovary	Endometriosis	666
Mammary gland (“blue domed”) simple cyst	Fibrocystic change of the breast	667
Eosinophilic cytoplasmic inclusions in Leydig cells	Reinke crystals (Leydig cell tumor)	671
Interdigitating layers of pink and red in arterial thrombi	Lines of Zahn (layers of platelets and RBCs seen only in thrombi formed before death)	691
Eosinophilic, hexagonal, double-pointed crystals in bronchial secretions	Charcot-Leyden crystals (asthma)	693
Whorled mucus plugs formed from shed bronchial epithelium	Curschmann spirals (asthma)	693
“Honeycomb” appearance of the lung on CXR or CT	Idiopathic pulmonary fibrosis	694
Golden-brown fusiform rods resembling dumbbells in alveolar sputum, visualized with Prussian blue stain	Asbestos (ferruginous) bodies	696
Bronchogenic apical lung tumor on imaging	Pancoast (superior sulcus) tumor (can compress cervical sympathetic chain → Horner syndrome)	704

► KEY ASSOCIATIONS

DISEASE/FINDING	MOST COMMON/IMPORTANT ASSOCIATIONS	PAGE
Mitochondrial inheritance	Disease occurs in all offspring of affected females (maternal inheritance pattern), heteroplasmy	55, 57
Intellectual disability	Down syndrome (sporadic), fragile X syndrome (inherited)	60, 61
Vitamin deficiency (USA)	Folate (pregnant women are at ↑ risk; body stores only a 3–4 month supply)	66
Lysosomal storage disease	Gaucher disease	86
HLA-DR3	Type I DM, SLE, Graves disease, Hashimoto thyroiditis, Addison disease	98
HLA-DR4	Rheumatoid arthritis, type I DM, Addison disease	98
Bacteria associated with gastritis, peptic ulcer disease, and gastric malignancies (eg, adenocarcinoma, MALToma)	<i>H pylori</i>	144
Opportunistic respiratory infection in AIDS	<i>Pneumocystis jirovecii</i>	151
Viral encephalitis affecting temporal lobe	HSV-1	162
Viral infection 2° to blood transfusion	Hepatitis C	171
Food poisoning (exotoxin-mediated)	<i>S aureus</i> , <i>B cereus</i>	174
Healthcare-associated pneumonia	<i>S aureus</i> , <i>Pseudomonas</i> , other enteric gram ⊖ rods	175
Bacterial meningitis (0–6 months old)	Group B streptococcus, <i>E coli</i> , <i>Listeria</i>	176
Bacterial meningitis (> 6 months old)	<i>S pneumoniae</i>	176
Osteomyelitis	<i>S aureus</i> (most common overall)	176
Osteomyelitis in sickle cell disease	<i>Salmonella</i> , <i>S aureus</i>	176
Osteomyelitis with injection drug use	<i>S aureus</i> , <i>Pseudomonas</i> , <i>Candida</i>	176
UTI	<i>E coli</i> , <i>Staphylococcus saprophyticus</i>	178
Bacterial STI	<i>C trachomatis</i> (D–K)	179
Pelvic inflammatory disease (PID)	<i>C trachomatis</i> (subacute), <i>N gonorrhoeae</i> (acute)	181
Metastases to bone	Prostate, breast >> lung > kidney, colon	219
Metastases to liver	Colon > breast >> pancreas, lung, prostate	219
Metastases to brain	Lung > breast >> melanoma > colon, prostate	219
S3 heart sound	↑ ventricular filling pressure (eg, MR, AR, HF, thyrotoxicosis), common in dilated ventricles	292
S4 heart sound	Stiff/hypertrophic ventricle (aortic stenosis, restrictive cardiomyopathy)	292
Holosystolic murmur	VSD, tricuspid regurgitation, mitral regurgitation	296
Ejection click	Aortic stenosis	296
Mitral stenosis	Rheumatic heart disease (late and highly specific sequelae of rheumatic fever)	296
Opening snap	Mitral stenosis	296
Heart murmur, congenital	Mitral valve prolapse	296

DISEASE/FINDING	MOST COMMON/IMPORTANT ASSOCIATIONS	PAGE
Cyanotic heart disease (early)	Tetralogy of Fallot (most common), D-transposition of great arteries, persistent truncus arteriosus, total anomalous pulmonary venous return, tricuspid atresia (right-to-left shunts)	302
Congenital heart disease (left-to-right shunts)	VSD > ASD > PDA	303
Late cyanotic shunt (uncorrected left-to-right shunt becomes right-to-left)	Eisenmenger syndrome (caused by VSD, ASD, PDA)	303
2° hypertension	Renal/renovascular diseases (eg, fibromuscular dysplasia, atherosclerotic renal artery stenosis), 1° hyperaldosteronism, OSA	304
Sites of atherosclerosis	Abdominal aorta > coronary artery > popliteal artery > carotid artery > circle of Willis	305
Aortic aneurysm, thoracic	Marfan syndrome (cystic medial degeneration), 3° syphilis (obliterative endarteritis of vasa vasorum)	306
Aortic aneurysm, abdominal	Atherosclerosis, tobacco use	306
Aortic dissection	Hypertension (most important risk factor)	307
Irregularly irregular rhythm on ECG with no discrete P waves	Atrial fibrillation (associated with ↑ risk of emboli)	311
Right heart failure due to a pulmonary cause	Cor pulmonale	316
Heart valve in infective endocarditis	Mitral > aortic, tricuspid (injection drug use)	318
Infective endocarditis presentation associated with bacteria	<i>S aureus</i> (acute, injection drug use, tricuspid valve), viridans streptococci (subacute, dental procedure), <i>S gallolyticus</i> (colon cancer), gram ⊖ (HACEK), culture ⊖ (<i>Coxiella</i> , <i>Bartonella</i>)	318
Cardiac tumor (adults)	Metastasis, myxoma (75%–80% in left atrium; “ball valve”)	320
Cardiac 1° tumor (children)	Rhabdomyoma (associated with tuberous sclerosis)	320
Hypercoagulability, endothelial damage, blood stasis	Virchow triad (↑ risk of thrombosis)	321
Congenital adrenal hyperplasia	21-hydroxylase deficiency	339
Hypopituitarism	Pituitary adenoma (under-secretion due to mass effect)	343
Congenital hypothyroidism	Thyroid dysgenesis/dyshormonogenesis, iodine deficiency	345
Thyroid cancer	Papillary carcinoma (<i>RET/PTC</i> rearrangements, <i>BRAF</i> mutations, childhood irradiation)	347
Hypoparathyroidism	Accidental excision during thyroidectomy	348
1° hyperparathyroidism	Adenomas, hyperplasia	349
2° hyperparathyroidism	Hypocalcemia of chronic kidney disease	349
Cushing syndrome (↑ cortisol)	Exogenous glucocorticoids; adrenocortical adenoma (secretes excess cortisol); ACTH-secreting pituitary adenoma (Cushing disease); paraneoplastic (due to ACTH secretion by tumors)	352
Cushing disease	↓ ACTH and ↓ cortisol in high-dose dexamethasone suppression test, ↑ ACTH and ↑ cortisol in CRH stimulation test	352
1° hyperaldosteronism	Bilateral adrenal hyperplasia or adenoma (Conn syndrome)	354
Tumor of the adrenal medulla (children)	Neuroblastoma (malignant)	354
Tumor of the adrenal medulla (adults)	Pheochromocytoma (usually benign)	355

DISEASE/FINDING	MOST COMMON/IMPORTANT ASSOCIATIONS	PAGE
Refractory peptic ulcers and ↑ gastrin levels (even after the secretin test)	Zollinger-Ellison syndrome (due to gastrin-secreting tumor of the duodenum or pancreas), associated with MEN1	357
Esophageal cancer	Squamous cell carcinoma (worldwide); adenocarcinoma (US)	385
Acute gastric ulcer associated with CNS injury	Cushing ulcer (↑ vagal stimulation → ↑ ACh → ↑ H ⁺ production)	386
Acute gastric ulcer associated with severe burns	Curling ulcer (hypovolemia → mucosal ischemia)	386
Chronic atrophic gastritis	↑ risk of gastric cancers, pernicious anemia (if autoimmune)	386
Bilateral ovarian metastases from gastric carcinoma	Krukenberg tumor (mucin-secreting signet ring cells)	386
Alternating areas of transmural inflammation and normal colon	Skip lesions (Crohn disease)	389
Site of diverticulosis	Sigmoid colon	390
False pharyngoesophageal diverticulum	Zenker diverticulum	391
Hepatocellular carcinoma	HBV (+/- cirrhosis), other causes of cirrhosis (HCV, alcoholic liver disease), specific carcinogens (eg, aflatoxins)	399
Inherited conjugated hyperbilirubinemia 2° to hepatocyte inability to secrete conjugated bilirubin in bile	Dubin-Johnson syndrome (black liver), Rotor syndrome (uncolored liver)	401
Inherited benign unconjugated hyperbilirubinemia	Gilbert syndrome	401
Inherited <i>ATP7B</i> mutation (copper buildup in liver, brain, cornea [Kayser-Fleischer rings], kidneys)	Wilson disease	402
Multiple blood transfusions or hereditary <i>HFE</i> mutation (can result in heart failure, “bronze diabetes,” and ↑ risk of hepatocellular carcinoma)	Hemochromatosis	402
Pancreatitis (acute)	Gallstones, alcohol	404
Pancreatitis (chronic)	Alcohol (adults), cystic fibrosis (children)	404
Microcytic anemia	Iron deficiency, thalassemia, lead poisoning, sideroblastic anemia	424, 425
Autosplenectomy (fibrosis and shrinkage), Howell-Jolly bodies	Sickle cell anemia (hemoglobin S)	428
Inherited platelet disorder with GpIb deficiency	Bernard-Soulier syndrome (↓ platelet-to-vWF adhesion)	432
Inherited platelet disorder with GpIIb/IIIa deficiency	Glanzmann thrombasthenia (↓ platelet-to-platelet aggregation, defective platelet plug formation)	432
Hereditary thrombophilia commonly associated with recurrent pregnancy loss	Factor V Leiden (mutant factor V resistant to degradation)	433
Disseminated Intravascular Coagulopathy (DIC)	Heat stroke, snake bite, sepsis, trauma, obstetric complications, acute pancreatitis, malignancy, nephrotic syndrome, transfusion	433
Common malignancy associated with noninfectious fever and bimodal age distribution	Hodgkin lymphoma	434
Type of Hodgkin lymphoma (most common)	Nodular sclerosis	434
t(14;18)	Follicular lymphoma (<i>BCL-2</i> activation, anti-apoptotic oncogene)	435, 439

DISEASE/FINDING	MOST COMMON/IMPORTANT ASSOCIATIONS	PAGE
t(8;14)	Burkitt lymphoma (<i>c-myc</i> fusion, transcription factor oncogene)	435, 439
Type of non-Hodgkin lymphoma (most common in adults)	Diffuse large B-cell lymphoma	435
Age ranges for patient with ALL/CLL/AML/CML	ALL: child, CLL: adult > 60, AML: adult ~ 65, CML: adult 45–85	437
t(9;22)	Philadelphia chromosome, CML (<i>BCR-ABL</i> oncogene, tyrosine kinase activation), more rarely associated with ALL	437, 439
Vertebral compression fracture	Osteoporosis	467
HLA-B27	Psoriatic arthritis, ankylosing spondylitis, IBD-associated arthritis, reactive arthritis	475
Death in SLE	Renal disease (most common), infections, cardiovascular disease (accelerated CAD)	476
Giant cell arteritis	Risk of ipsilateral blindness due to occlusion of ophthalmic artery; associated with polymyalgia rheumatica	478
Recurrent inflammation/thrombosis of medium-sized vessels in extremities	Buerger disease (strongly associated with tobacco smoking, Raynaud phenomenon)	478
Benign vascular tumor of infancy	Infantile hemangioma (grows rapidly then involutes starting at age 1)	486
Herald patch (followed by scaly erythematous plaques in a “Christmas tree” distribution)	Pityriasis rosea	492
Actinic keratosis	Precursor to squamous cell carcinoma	494
Cerebellar tonsillar herniation	Chiari I malformation (associated with spinal cord cavitations [eg, syringomyelia])	502
Bilateral mamillary body lesions with thiamine deficiency	Wernicke-Korsakoff syndrome	524
Epidural hematoma	Rupture of middle meningeal artery (trauma; biconvex/lentiform-shaped)	528
Subdural hematoma	Rupture of bridging veins (trauma, cerebral atrophy; crescent-shaped)	528
Dementia	Alzheimer disease, vascular dementia (multiple infarcts, stepwise decline)	534, 535
Demyelinating disease in young women	Multiple sclerosis	537
Brain tumor (adults)	Metastasis, glioblastoma (malignant), meningioma, hemangioblastoma	540
Galactorrhea, amenorrhea	Prolactinoma	540
Brain tumor (children)	Overall: pilocytic astrocytoma (benign) Infratentorial: medulloblastoma (most common malignant) Supratentorial: craniopharyngioma (malignant)	542
Combined UMN and LMN degeneration	Amyotrophic lateral sclerosis	544
Degeneration of dorsal column fibers	Tabes dorsalis (3° syphilis), subacute combined degeneration (Vitamin B ₁₂ deficiency; dorsal columns, lateral corticospinal, spinocerebellar tracts affected)	544
Nephrotic syndrome (children)	Minimal change disease	616

DISEASE/FINDING	MOST COMMON/IMPORTANT ASSOCIATIONS	PAGE
Kidney stones (radiolucent)	Uric acid	617
Kidney stones (radiopaque)	Calcium (most common), struvite (ammonium), cystine (faintly radiopaque)	617
Renal malignancy (in males)	Renal cell carcinoma: associated with tobacco smoking and VHL (clear cell subtype); may present with paraneoplastic syndromes (EPO, renin, PTHrP, ACTH)	623
1° amenorrhea	Turner syndrome (45,XO or 45,XO/46,XX mosaic)	655
Hypogonadotropic hypogonadism with anosmia	Kallmann syndrome (neuron migration failure)	656
Clear cell adenocarcinoma of the vagina	DES exposure in utero	662
Ovarian tumor (benign, bilateral)	Serous cystadenoma	664
Ovarian tumor (malignant)	Serous carcinoma	664
Benign tumor of myometrium	Leiomyoma (estrogen sensitive, not precancerous)	666
Gynecologic malignancy (most common)	Endometrial carcinoma (most common in resource-rich countries); cervical cancer (most common worldwide)	663–666
Breast mass	Fibrocystic change (in premenopausal females); carcinoma (in postmenopausal females)	667, 668
Breast tumor (benign, young woman)	Fibroadenoma	667
Breast cancer	Invasive ductal carcinoma	668
Bloody nipple discharge	Intraductal papilloma (no palpable mass); invasive ductal carcinoma (hard palpable mass)	
Testicular tumor	Seminoma (malignant, radiosensitive, ↑ placental alkaline phosphatase [PLAP])	670, 671
Bladder outlet obstruction in men	BPH	672
Pulmonary hypertension	Idiopathic, left heart disease, lung disease/hypoxia, chronic thromboembolism, multifactorial	698
SIADH	Small cell carcinoma of the lung	703

► EQUATION REVIEW

TOPIC	EQUATION	PAGE
Volume of distribution	$V_d = \frac{\text{amount of drug in the body}}{\text{plasma drug concentration}}$	229
Half-life	$t_{1/2} = \frac{0.7 \times V_d}{CL}$	229
Drug clearance	$CL = \frac{\text{rate of elimination of drug}}{\text{plasma drug concentration}} = V_d \times K_e$ (elimination constant)	229
Loading dose	$LD = \frac{C_p \times V_d}{F}$	229
Maintenance dose	$\text{Maintenance dose} = \frac{C_p \times CL \times \tau}{F}$	229
Therapeutic index	$TI = \text{median toxic dose/median effective dose} = TD_{50}/ED_{50}$	233